

Irrationality of the 5-adic zeta value $\zeta_5(3)$

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Abstract

We prove that the Kubota–Leopoldt value $\zeta_5(3)$ is irrational. Our proof combines a level-five overconvergent Eisenstein germ, an explicit third-order modular differential equation of exact inhomogeneous order, Buzzard’s finite-slope continuation theorem, and an adelic arithmetic-holonomy bound. We prove the required row-wise adelic bound and evaluate the complex logarithmic energy exactly by Jensen’s formula on $X_0(5)$. The resulting incompatible estimates are

$$4 \leq \frac{I + 3 \log 5}{-\log 2 + 3 \log 5 - 45/16} = 3.926639729 \dots < 4.$$

Use of AI: We prepared a list of proof strategies and techniques used in the irrationality proofs of zeta values and p -adic zeta values. We fed this to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) and prompted these AI tools to prove irrationality of $\zeta(n)$ and $\zeta_p(n)$ for several values of n, p and also subsets of zeta values (to prove that at least one of the values in the subset is irrational). When an AI tool was hallucinating, stuck, or going in the wrong direction, we stopped that AI tool, corrected the intermediate .md files, added more specific new ideas and switched to a different AI tool with more detailed fresh prompts. Equations written by one AI tool were critiqued, verified and expanded by a completely different AI tool in an iterative manner, till we were satisfied with their correctness and details. Irrationality of $\zeta_5(3)$ is the first result we obtained in this domain. The certificates code was written by Codex and verified by the author. The author is solely responsible for the correctness of the proofs and the code.

1 Introduction

We use Calegari’s weight-space normalization [5]: $\zeta_5(3)$ is twice the constant term of the overconvergent Eisenstein series at the character $x \mapsto x^{-2}$ of $\mathbf{Z}_5^\times$. Equivalently,

$$\zeta_5(3) = \lim_{\substack{k \in \mathbf{Z}_{>0}, \ k \rightarrow +\infty \\ k \rightarrow 1 \text{ in } \mathbf{Z}_5 \\ 2k \equiv 2 \pmod{4}}} \zeta(1 - 2k).$$

Thus k tends to infinity in the usual sense while tending to 1 5-adically; the congruence condition keeps the sequence on the same component of weight space. This branch specification is essential because the Kubota–Leopoldt zeta function is naturally a rigid-analytic function on weight space, which has several components. We regard $\mathbf{Q}$ as a subfield of $\mathbf{Q}_5$; an element of $\mathbf{Q}_5$ is called irrational if it does not lie in $\mathbf{Q}$.

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Theorem 1.1. $\zeta_5(3) \in \mathbf{Q}_5 \setminus \mathbf{Q}$.

For Kubota–Leopoldt zeta values at specified positive odd arguments, the first irrationality results were obtained for the small primes. Calegari proved that $\zeta_2(3)$ and $\zeta_3(3)$ are irrational by means of overconvergent modular forms [5]. Beukers subsequently gave alternative continued-fraction proofs of these small-prime zeta-value results and treated further p -adic L -values and p -adic Hurwitz zeta values [1]. More recently, Calegari–Dimitrov–Tang proved the irrationality of the individual value $\zeta_2(5)$ [7]. A different Apéry-style proof, together with an explicit upper bound for its irrationality measure, was given by Lai–Sprang–Zudilin [11].

There are also substantial simultaneous results. Sprang proved, for every prime p , a p -adic analogue of the Ball–Rivoal theorem: the dimension of the rational span of the branchwise positive p -adic zeta values up to height s grows at least logarithmically with s [12]. For $p = 2$, Lai showed that, for every integer $s \geq 0$, the interval $[s + 3, 3s + 5]$ contains an odd j for which $\zeta_2(j)$ is irrational [8]. Lai and Sprang later proved that, for every prime p , there is a constant $c_p > 0$ such that, for every $\epsilon > 0$, at least $(c_p - \epsilon)\sqrt{s/\log s}$ of $\zeta_p(3), \zeta_p(5), \dots, \zeta_p(s)$ are irrational for all sufficiently large positive odd integers s [10]. Finally, Lai–Lupu–Sprang proved that, for every $p \geq 5$, there is an odd integer i with

$$3 \leq i \leq p + \frac{p}{\log p} + 5$$

for which $\zeta_p(i)$ is irrational [9]. For $p \geq 5$, these simultaneous results do not determine whether the specified value $\zeta_p(3)$ is irrational. Calegari’s direct level-five approximants yield the exponent

$$\frac{3}{3/2 + 3/\log 5} = 0.891794\dots < 1,$$

and hence fall short of the irrationality criterion [5]. Theorem 1.1 supplies the missing individual conclusion for $p = 5$.

Outline the proof: In Section 2, we introduce the level-five coordinate f , the modular form A , and the overconvergent Eisenstein germ $H(f) = A(q(f))(G(q(f)) + \zeta_5(3)/2)$. Under the contrary assumption $\zeta_5(3) \in \mathbf{Q}$, we have $H \in \mathbf{Q}[[f]]$. Section 3, proves a row-wise adelic holonomy bound that converts coefficient-denominator estimates and adelic meromorphic-continuation data into an upper bound for the number of linearly independent holonomic germs. Its column-by-column denominator optimization is essential for the final constant. In Section 4, we derive an explicit third-order inhomogeneous differential equation for H and prove that its order is minimal. With $\delta = f d/df$, this gives the linear independence of $1, H, \delta H, \delta^2 H$ over $\mathbf{Q}(f)$. Section 5, proves the three denominator bounds of type D_n^3 , where $D_n = \text{lcm}(1, \dots, n)$, and uses Buzzard’s finite-slope continuation theorem to extend the three nonconstant germs to the strict disk $|f|_5 < 125$. Section 6, evaluates the archimedean boundary energy exactly by Jensen’s formula on $X_0(5)$. Finally, Section 7, applies the holonomy bound to $1, H, \delta H, \delta^2 H$: the denominator rows give the optimized invariant $45/16$, and the local maps and energy calculation force the resulting quotient to be at least four. The rational numerical certificate proves that the same quotient is strictly smaller than four, yielding the required contradiction.

2 The level-five Eisenstein germ

Let $\tau \in \mathfrak{H}$, put $q = e^{2\pi i\tau}$, and write $\theta = q d/dq$. We use the conventions $\Delta = \eta^{24}$ and $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$. Define

$$f = f_5(\tau) = \left(\frac{\Delta(5\tau)}{\Delta(\tau)} \right)^{1/4} = \frac{\eta(5\tau)^6}{\eta(\tau)^6} = q \prod_{n \geq 1} \left(\frac{1 - q^{5n}}{1 - q^n} \right)^6. \quad (1)$$

Here the fourth root is the one normalized by $f = q + O(q^2)$. For the eta-quotient exponents $r_1 = -6$ and $r_5 = 6$, we have

$$\sum_{d|5} dr_d = 24, \quad \sum_{d|5} \frac{5}{d} r_d = -24, \quad \prod_{d|5} d^{r_d} = 5^6.$$

The eta-quotient criterion therefore shows that f is a weight-zero modular function on $\Gamma_0(5)$ with trivial character.

Let $w_5\tau = -1/(5\tau)$. The transformation formula $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ gives

$$f(w_5\tau) = \frac{\eta(-1/\tau)^6}{\eta(-1/(5\tau))^6} = \frac{1}{125f(\tau)}. \quad (2)$$

The product in (1) shows that f has a simple zero at ∞ , while (2) shows that it has a simple pole at the cusp 0. The function η has no zeros on $\mathfrak{H}$, and these are the only two cusps of $X_0(5)$. Consequently

$$\operatorname{div}(f) = (\infty) - (0).$$

In particular, the map $f : X_0(5) \rightarrow \mathbf{P}^1$ has degree one; hence it is an isomorphism and f is a global coordinate on $X_0(5)$.

Since

$$\frac{1 - q^{5n}}{1 - q^n} = 1 + q^n + q^{2n} + q^{3n} + q^{4n},$$

we have $f \in q + q^2\mathbf{Z}[[q]]$. Recursive formal inversion of a series with linear coefficient one takes place over $\mathbf{Z}$, so $q(f) \in f + f^2\mathbf{Z}[[f]]$.

Define

$$A(\tau) = \frac{5E_2(5\tau) - E_2(\tau)}{4} = \frac{\theta f}{f} = 1 + 6 \sum_{n \geq 1} \sum_{\substack{d|n \\ 5 \nmid d}} d q^n, \quad (3)$$

$$G(q) = \sum_{n \geq 1} \left(\sum_{\substack{d|n \\ 5 \nmid d}} d^{-3} \right) q^n, \quad (4)$$

$$\eta_5 = \frac{\zeta_5(3)}{2}. \quad (5)$$

Indeed, $\theta\eta/\eta = E_2/24$ proves the first two expressions for A . Using $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n)q^n$, its coefficient of q^n is

$$6(\sigma_1(n) - 5\sigma_1(n/5)) = 6 \sum_{\substack{d|n \\ 5 \nmid d}} d,$$

where $\sigma_1(n/5) = 0$ when $5 \nmid n$. This proves the last expression in (3). Equivalently, since f is a modular unit, its logarithmic derivative $A = \theta f/f$ is a holomorphic classical modular form of weight two on $\Gamma_0(5)$.

Calegari's overconvergent Eisenstein family [5] specializes at the character $x \mapsto x^{-2}$ to

$$E_{-2}^{*(5)} = \eta_5 + G.$$

It is an overconvergent form of weight -2 . With the normalization

$$U_5 \left(\sum_{n \geq 0} a_n q^n \right) = \sum_{n \geq 0} a_{5n} q^n,$$

it has eigenvalue one: the divisors of $5n$ that are not divisible by five are precisely the divisors of n that are not divisible by five, and hence

$$\sum_{\substack{d|5n \\ 5 \nmid d}} d^{-3} = \sum_{\substack{d|n \\ 5 \nmid d}} d^{-3}.$$

The constant term η_5 is also fixed by U_5 . Thus this eigenform has slope zero.

The product

$$H(f) = A(q(f))(G(q(f)) + \eta_5) \tag{6}$$

is therefore the germ at $f = 0$ of a weight-zero overconvergent modular function. In the remainder of the proof we assume, for contradiction, that $\zeta_5(3) \in \mathbf{Q}$. Then $\eta_5 \in \mathbf{Q}$; since $A, G \in \mathbf{Q}[[q]]$ and $q(f) \in \mathbf{Z}[[f]]$, equation (6) implies $H \in \mathbf{Q}[[f]]$.

3 A row-wise adelic holonomy bound

We isolate the precise slopes result that will be applied. The archimedean theorem and its Bost–Charles metric are proved in [6]; the adelic metric and the finite-place evaluation estimates are developed in [7]. We include the deduction needed here, retaining the auxiliary denominator lattice because its parameters must be optimized one column at a time. Throughout this section, $|p|_p = p^{-1}$, and “holonomic” means that the series satisfies a nonzero linear differential equation over $\mathbf{Q}(x)$.

For $t \geq 0$, write

$$\Lambda(t) = \text{lcm}(1, 2, \dots, \lfloor t \rfloor), \quad \Lambda(t) = 1 \quad (0 \leq t < 1).$$

Let $\mathbf{T} = \{z \in \mathbf{C} : |z| = 1\}$, equipped with normalized Haar probability measure μ .

Theorem 3.1 (row-wise adelic holonomy bound). *Let $m, r \geq 1$, and let $F_1, \dots, F_m \in \mathbf{Q}[[x]]$ be holonomic and linearly independent over $\mathbf{Q}(x)$. Suppose their coefficient denominators are capped by an $m \times r$ matrix $\mathbf{b} = (b_{i,j})$ whose j -th column has the form*

$$b_{1,j} = \dots = b_{u_j,j} = 0, \quad b_{u_j+1,j} = \dots = b_{m,j} = b_j > 0,$$

where $0 \leq u_j \leq m$ and an empty string of equalities is omitted. More precisely, assume that there is an integer $C \geq 1$, independent of i and n , such that

$$CF_i(x) = \sum_{n \geq 0} \frac{a_{i,n} x^n}{\prod_{j=1}^r \Lambda(b_{i,j} n)}, \quad a_{i,n} \in \mathbf{Z}. \tag{7}$$

For each place v of $\mathbf{Q}$, let φ_v be analytic on a neighborhood of the closed unit disk over $\mathbf{Q}_v$ (over $\mathbf{C}$ when $v = \infty$), with $\varphi_v(0) = 0$ and $\varphi'_v(0) \neq 0$. Suppose that $\varphi_p(z) = z$ for all but finitely many

primes and that every $F_i \circ \varphi_v$ is meromorphic on the open unit disk. Put

$$\mathcal{A} = \iint_{\mathbf{T}^2} \log |\varphi_\infty(z) - \varphi_\infty(w)| d\mu(z) d\mu(w) + \sum_p \sup_{|z|_p=1} \log |\varphi_p(z)|_p, \quad (8)$$

$$\mathcal{L} = \log |\varphi_\infty'(0)| + \sum_p \log |\varphi_p'(0)|_p, \quad (9)$$

$$\tau(\mathbf{b}) = \sum j = 1^r b_j - \frac{1}{m^2} \sum *j = 1^r u_j^2 b_j. \quad (10)$$

All sums over finite primes are finite because $\varphi_p(z) = z$ almost everywhere. The double integral is finite: φ_∞ is nonconstant, and the zero set of $\varphi_\infty(z) - \varphi_\infty(w)$ produces only locally integrable logarithmic singularities on $\mathbf{T}^2$. If $\sigma_i = \sum_j b_{i,j}$ is the i -th row sum, then

$$\tau(\mathbf{b}) = \frac{1}{m^2} \sum_{i=1}^m (2i-1) \sigma_i, \quad 0 \leq \tau(\mathbf{b}) \leq \sum_{j=1}^r b_j. \quad (11)$$

Indeed, the contribution of column j to the middle expression is $m^{-2} b_j \sum_{i=u_j+1}^m (2i-1) = b_j(1 - u_j^2/m^2)$. If $\mathcal{L} > \tau(\mathbf{b})$, then

$$m \leq \frac{\mathcal{A}}{\mathcal{L} - \tau(\mathbf{b})}. \quad (12)$$

Proof. Replacing each F_i by CF_i does not affect any hypothesis or the conclusion. We may therefore take $C = 1$ in (7); a fixed scaling would in any event contribute only $O(D)$ to the sum of evaluation heights below.

Fix real parameters $0 \leq y_j \leq mb_j$. For an integer $D \geq 1$, let $\mathbf{Z}[1/x] * \leq D$ be the free module with basis $1, x^{-1}, \dots, x^{-D}$, and define the rank- $m(D+1)$ lattice

$$E_D = \bigoplus *i = 1^m \left(\prod_{j:i \leq u_j} \Lambda(y_j D)^{-1} \right) e_i \mathbf{Z}[1/x]_{\leq D}, \quad (13)$$

with evaluation map

$$\psi_D : E_D \otimes \mathbf{Q} \longrightarrow x^{-D} \mathbf{Q}[[x]], \quad e_i P(x) \longmapsto P(x) F_i(x).$$

This map is injective: an element of its kernel would be a nontrivial $\mathbf{Q}(x)$ -linear relation among the F_i .

We recall the metrics used in the cited adelic slopes construction. At the archimedean place, every polynomial block carries the L^2 -metric induced by the Bost–Charles metric attached to φ_∞ . At a finite prime p , put

$$s_p = \sup_{|z|_p=1} \log |\varphi_p(z)|_p$$

and give the basis vector x^{-k} the diagonal norm e^{-ks_p} . For all but finitely many primes, $s_p = 0$ and this is the standard norm. The arithmetic Hilbert–Samuel formula and the Bost–Charles self-intersection calculation in [3] give

$$\widehat{\deg} \overline{E}_D = \left(\frac{m}{2} \mathcal{A} + \sum j = 1^r u_j y_j \right) D^2 + o(D^2). \quad (14)$$

Indeed, if $\mathcal{A} * \infty$ denotes the double integral in equation (8), the archimedean contribution of one block is $\mathcal{A}_\infty D^2/2 + o(D^2)$. At p , the determinant of the diagonal metric contributes $s_p \sum_{k=0}^D k =$

$s_p D^2/2 + O(D)$ per block. Finally, $\Lambda(y_j D)^{-1}$ occurs in $u_j(D+1)$ lattice basis vectors, and $\log \Lambda(y_j D) = y_j D + o(D)$ by the prime number theorem. These three contributions give equation (14).

For $n \geq 0$, filter the target and the source by

$$\mathcal{F}_D^{(n)} = x^{n-D} \mathbf{Q}[[x]], \quad E_D^{(n)} = E_D \cap \psi_D^{-1}(\mathcal{F}_D^{(n)}).$$

Each graded piece $E_D^{(n)}/E_D^{(n+1)}$ has rank zero or one. Let

$$\mathcal{V}_D = \{n \geq 0 : \text{rank}(E_D^{(n)}/E_D^{(n+1)}) = 1\}.$$

The filtration is separated and exhaustive, so $|\mathcal{V}_D| = \text{rank } E_D = m(D+1)$. For $n \in \mathcal{V}_D$, evaluation induces an injection

$$\psi_D^{(n)} : E_D^{(n)}/E_D^{(n+1)} \longrightarrow \mathcal{F}_D^{(n)}/\mathcal{F}_D^{(n+1)}.$$

Write $h_v(\psi_D^{(n)})$ for its local height with respect to the metrics just specified and the norm-one generator x^{n-D} of the target.

The Chudnovsky–Osgood functional bad-approximability theorem for an independent holonomic system [6] implies that, for every $\epsilon > 0$, there is a constant C_ϵ such that

$$\mathcal{V}_D \subset \{0, 1, \dots, \lfloor (m + \epsilon)(D + 1) + C_\epsilon \rfloor\}. \quad (15)$$

The Bost–Charles subharmonic estimate at infinity and its nonarchimedean maximum-principle analogue at the finite places give, for $n \in \mathcal{V}_D$,

$$\sum_v h_v(\psi_D^{(n)}) \leq -n\mathcal{L} + D\mathcal{A} + \sum_{j=1}^r b_j \max \left\{ n, \frac{y_j}{b_j} D \right\} + o(n + D). \quad (16)$$

The hypothesis of meromorphy on the open disk is sufficient here. Restrict to a closed subdisk of radius $\rho < 1$, rescale it to the unit disk, clear the finitely many poles there by a common factor equal to one at the origin, and apply the closed-disk estimate. The clearing factor contributes only $O(1)$ to each local height. One first lets $D \rightarrow \infty$ and then $\rho \uparrow 1$; this is the standard limiting argument in the sources [6] and [7].

We spell out the denominator term, which is the only modification of the standard local estimates. Write $P_i = \sum_{k=0}^D c_{i,k} x^{-k}$. A contribution to the coefficient of x^{n-D} in $P_i F_i$ uses the coefficient of F_i with index $n - D + k$; whenever it occurs, this index lies between 0 and n . Thus, in column j , a row $i > u_j$ has coefficient denominator dividing $\Lambda(b_j n)$, whereas a row $i \leq u_j$ has no denominator from F_i in that column but has the lattice denominator $\Lambda(y_j D)$. The least common multiple over all rows divides $\Lambda(\max\{b_j n, y_j D\})$, because the smaller of two initial-segment least common multiples divides the larger. Summing its valuations over the finite primes and using the prime number theorem contributes $b_j \max\{n, (y_j/b_j)D\} + o(n + D)$. The analytic parts of the local estimates supply $-n\mathcal{L} + D\mathcal{A}$; see [6] and [7]. For fixed ϵ and fixed $y_1, \dots, y_r$, the error in equation (16) is uniform for the range of n in (15).

We now sum the local estimates. Since $\mathcal{L} > \tau(\mathbf{b}) \geq 0$, the coefficient of n in the first term is negative. If $N_D = m(D+1)$, the jumps are distinct nonnegative integers, and therefore

$$\sum_{n \in \mathcal{V}_D} n \geq \frac{N_D(N_D - 1)}{2} = \frac{m^2}{2} D^2 + o(D^2). \quad (17)$$

For the increasing function $g_j(t) = b_j \max\{t, y_j/b_j\}$, the largest sum over N_D distinct integers in the interval from (15) uses its N_D largest integers. After division by D^2 , this is an upper Riemann sum on $[\epsilon, m + \epsilon]$, up to $o(1)$. Letting $D \rightarrow \infty$ and then $\epsilon \downarrow 0$ gives

$$\limsup_{D \rightarrow \infty} \frac{1}{D^2} \sum_{n \in \mathcal{V}_D} b_j \max\left\{n, \frac{y_j}{b_j} D\right\} \leq b_j \int_0^m \max\left\{t, \frac{y_j}{b_j}\right\} dt. \quad (18)$$

The uniform error in (16) sums to $o(D^2)$. In the present rank-one graded setting, Bost's slopes inequality is

$$\widehat{\deg} \overline{E}_D \leq \sum_{n \in \mathcal{V}_D} \sum_v h_v(\psi_D^{(n)});$$

see [2]. Combining this inequality with equations (17) and (18) yields

$$\widehat{\deg} \overline{E}_D \leq \left(-\frac{m^2}{2} \mathcal{L} + m\mathcal{A} + \sum_{j=1}^r b_j \int_0^m \max\left\{t, \frac{y_j}{b_j}\right\} dt \right) D^2 + o(D^2). \quad (19)$$

Here the term $m\mathcal{A}D^2$ is simply $|\mathcal{V}_D|D\mathcal{A} + o(D^2)$. Since $0 \leq y_j/b_j \leq m$,

$$b_j \int_0^m \max\left\{t, \frac{y_j}{b_j}\right\} dt = \frac{1}{2} \left(m^2 b_j + \frac{y_j^2}{b_j} \right),$$

comparison of (14) and (19) yields

$$m^2 \left(\mathcal{L} - \sum_j b_j \right) \leq m\mathcal{A} + \sum_j \left(\frac{y_j^2}{b_j} - 2u_j y_j \right). \quad (20)$$

The right side is minimized at $y_j = u_j b_j$, which lies in the allowed range $[0, mb_j]$. Substitution in (20) gives

$$m^2 \left(\mathcal{L} - \sum_j b_j \right) \leq m\mathcal{A} - \sum_j u_j^2 b_j,$$

or equivalently $m(\mathcal{L} - \tau(\mathbf{b})) \leq \mathcal{A}$. The assumed strict positivity of $\mathcal{L} - \tau(\mathbf{b})$ now permits division and proves equation (12). $\square$

Remark 3.2. The crucial feature is the subtraction $m^{-2} \sum_j u_j^2 b_j$ in $\tau(\mathbf{b})$. It comes from minimizing the quadratic in (20) before the columns are combined. The same row-wise optimization is used in the adelic application to $\zeta_2(5)$ in [7].

4 The differential equation and its exact order

Put $\delta = f d/df$ and

$$\begin{aligned} P &= 1 + 22f + 125f^2, & Q &= 1 + 23f + 125f^2, & T &= 1 + 24f + 125f^2, \\ N &= 1 + 250f + 3125f^2, & M &= 1 + 10f + 5f^2. \end{aligned}$$

We first establish the modular identities used below. At the cusp ∞ , the functions $j(\tau)$ and $j(5\tau)$ have pole orders one and five, respectively. At the cusp 0, their pole orders are five and one,

respectively: indeed, $j(w_5\tau) = j(5\tau)$ and $j(5w_5\tau) = j(\tau)$. Since f has divisor $(\infty) - (0)$, the functions $j(\tau)f$ and $j(5\tau)f^5$ are regular away from 0 and have there a pole of order at most six. They are therefore polynomials in f of degree at most six. Exact formal expansion through degree six at $f = 0$, obtained from (1), $j = E_4^3/\Delta$, and the recursively determined $q(f)$, gives

$$j(\tau)f = N^3 + O(f^7), \quad j(5\tau)f^5 = M^3 + O(f^7).$$

The degree bound makes this finite comparison conclusive. Thus the Hauptmodul identities, in the normalization (1), are

$$j(\tau) = \frac{N^3}{f}, \quad j(5\tau) = \frac{M^3}{f^5}. \quad (21)$$

For a direct derivation of the weight-four identities, put

$$R_1 = 15625f^2 + 500f - 1, \quad R_2 = f^2 - 4f - 1.$$

Logarithmic differentiation of (21) gives

$$\delta \log j(\tau) = \frac{R_1}{N}, \quad \frac{1}{5} \delta \log j(5\tau) = \frac{R_2}{M}.$$

Using $\theta = A\delta$ and the identities

$$\frac{\theta j(\tau)}{j(\tau)} = -\frac{E_6(\tau)}{E_4(\tau)}, \quad \frac{\theta j(5\tau)}{j(5\tau)} = -5\frac{E_6(5\tau)}{E_4(5\tau)}, \quad \frac{E_6^2}{E_4^2} = E_4 \left(1 - \frac{1728}{j}\right),$$

together with the exact polynomial factorizations

$$\begin{aligned} N^3 - 1728f &= PR_1^2, \\ M^3 - 1728f^5 &= PR_2^2, \end{aligned} \quad (22)$$

one obtains

$$\frac{E_4(\tau)}{A^2} = \frac{N}{P}, \quad \frac{E_4(5\tau)}{A^2} = \frac{M}{P}, \quad \frac{E_4(\tau) - E_4(5\tau)}{240} = A^2 \frac{f(1 + 13f)}{P}. \quad (23)$$

For example, the first identity is the exact simplification

$$\frac{E_4(\tau)}{A^2} = \frac{(R_1/N)^2}{1 - 1728f/N^3} = \frac{N}{P};$$

the second is identical with R_2, M . The third follows from the first two and $N - M = 240f(1 + 13f)$. All comparisons above take place in $\mathbf{Z}[[q]]$ or $\mathbf{Q}(f)$; no numerical recognition is involved.

Define

$$\mathcal{L}_3 = P^3\delta^3 - 12fPQ\delta + 6f(125f^2 - 1)T. \quad (24)$$

Lemma 4.1 (exact modular equation). *For every thrice differentiable germ $g(\tau)$,*

$$\mathcal{L}_3(Ag) = \frac{P^3}{A^2} \theta^3 g. \quad (25)$$

Consequently,

$$\mathcal{L}_3 H = f(1 + 13f)P^2. \quad (26)$$

Proof. Let $D = \delta$, $u = DA/A$, and $s = D\tau = (2\pi iA)^{-1}$, so $Ds/s = -u$. With $z = 1728/j = 1728f/N^3$, the inverse uniformizing map is the ratio of two solutions, up to a fractional-linear transformation, of the hypergeometric equation

$$z(1-z)y'' + \left(1 - \frac{3}{2}z\right)y' - \frac{5}{144}y = 0.$$

The standard Schwarzian formula for a ratio of solutions therefore gives

$$\{\tau, z\} = \frac{32z^2 - 41z + 36}{72z^2(1-z)^2}.$$

Here $\{h, x\} = h'''/h' - \frac{3}{2}(h''/h')^2$ denotes the Schwarzian derivative, with derivatives taken with respect to x . The Schwarzian chain rule gives $\{\tau, \log f\} = \{\tau, z\}(Dz)^2 + \{z, \log f\}$. Substitution of $z = 1728f/N^3$ and exact simplification yield the first identity below. On the other hand, $D\tau = s$ and $Ds = -us$ give $\{\tau, \log f\} = -Du - u^2/2$, so

$$\{\tau, \log f\} = -\frac{6fQ}{P^2}, \quad 2Du + u^2 = \frac{12fQ}{P^2}. \quad (27)$$

Differentiating and using

$$D\left(\frac{fQ}{P^2}\right) = \frac{f(1-125f^2)T}{P^3}$$

gives $D^2u + uDu = 6f(1-125f^2)T/P^3$. Also,

$$\frac{D^3A}{A} = D^2u + 3uDu + u^3 = (D^2u + uDu) + u(2Du + u^2). \quad (28)$$

Writing primes for $d/d\tau$, three applications of the product rule give $D^3(Ag) = (D^3A)g + As(2Du + u^2)g'(\tau) + As^3g'''(\tau)$; the g'' -coefficient cancels because $Ds = -us$. Substitution in (24), using (27) and (28), cancels the g - and g' -terms. Finally, $As^3g''' = A^{-2}\theta^3g$, so the remaining term is $P^3A^{-2}\theta^3g$, proving (25).

Finally, coefficientwise differentiation of (4) gives

$$\theta^3G = \frac{E_4(\tau) - E_4(5\tau)}{240}.$$

Indeed, the coefficient of q^n on both sides is $\sum_{d|n, 5 \nmid d} (n/d)^3$. Equations (23) and (25), first with $g = 1$ and then with $g = G$, prove (26). $\square$

Definition 4.2. The *inhomogeneous differential order* of a germ Y over $\mathbf{C}(f)$ is the least integer $r \geq 0$ for which

$$\delta^r Y + a_{r-1}\delta^{r-1}Y + \cdots + a_0Y = b, \quad a_i, b \in \mathbf{C}(f).$$

For $r = 0$, this means simply $Y = b$. If no such r exists, the order is infinite. Equivalently, when the order is finite, it is one less than the dimension of the $\mathbf{C}(f)$ -span of $1, Y, \delta Y, \dots$: the first dependence in this ordered list is precisely a normalized inhomogeneous equation of minimal order.

Proposition 4.3 (exact inhomogeneous order). *For every $c \in \mathbf{C}$, the germ $A(G + c)$ has exact inhomogeneous order three over $\mathbf{C}(f)$. In particular,*

$$1, H, \delta H, \delta^2 H$$

are linearly independent over $\mathbf{Q}(f)$ under the standing assumption $\eta_5 \in \mathbf{Q}$.

Proof. Fix $c \in \mathbf{C}$ and put $Y_c = A(G + c)$. Since $\mathcal{L}_3 A = 0$, Lemma 4.1 gives

$$\mathcal{L}_3 Y_c = f(1 + 13f)P^2, \quad (29)$$

so the inhomogeneous order is at most three.

Equation (25) shows that the local homogeneous solution space V of $\mathcal{L}_3$ is

$$A\langle 1, \tau, \tau^2 \rangle.$$

Under modular continuation this is the symmetric square of the standard two-dimensional representation. Its monodromy is irreducible: $\Gamma_0(5)$ has finite index in $\mathrm{SL}_2(\mathbf{Z})$, hence is Zariski dense in SL_2 , and Sym^2 of the standard representation is irreducible.

We next prove that equation (29) has no rational particular solution. Thus there is no $r(f) \in \mathbf{C}(f)$ satisfying

$$\mathcal{L}_3 r = f(1 + 13f)P^2. \quad (30)$$

The discriminant of P is -16 , so its two roots are distinct. At either root a , substitution of $(f - a)^\lambda$ into the lowest-order terms uses

$$Q(a) = a, \quad T(a) = 2a, \quad (aP'(a))^2 = -16a^2,$$

and the normalized indicial polynomial is

$$I_a(\lambda) = \lambda(\lambda - 1)(\lambda - 2) + \frac{3}{4}\lambda - \frac{3}{4} = \left(\lambda - \frac{1}{2}\right)(\lambda - 1)\left(\lambda - \frac{3}{2}\right). \quad (31)$$

The right side of (30) vanishes to order two at a . Since no negative integer is a root of (31), a rational solution cannot have a pole at either root of P .

At an ordinary finite point $b \neq 0$ (so $P(b) \neq 0$), the coefficient of the most singular term of a putative principal part $c_0(f - b)^{-k}$, $k > 0$, is $P(b)^3 b^3 c_0(-k)(-k - 1)(-k - 2)(f - b)^{-k-3}$, which is nonzero. At $f = 0$, a term $c_0 f^{-k}$ contributes $c_0(-k)^3 f^{-k}$ through $P^3 \delta^3$, while the other terms have strictly larger f -order. Thus a rational solution has no finite poles and must be a polynomial.

If its degree is $d > 0$, the unique term of degree $d + 6$ is the leading term of $P^3 \delta^3 r$, with nonzero coefficient $125^3 d^3$ times the leading coefficient of r . The other terms have degree at most $d + 5$, whereas the right side of (30) has degree six. This is impossible. If r is constant, the left side has degree at most five, again contradicting the degree-six right side. This proves the claim.

We now record the regular-singular property needed for the monodromy argument. The only singular points of the homogeneous equation on $\mathbf{P}_f^1$ are $0, \infty$, and the two roots of P . At a root of P , passage to the usual monic equation gives a holomorphic coefficient of the second derivative, a pole of order at most two in the coefficient of the first derivative, and a pole of order at most three in the zeroth-order coefficient. Hence these two points are regular singular; regular singularity at 0 is immediate from the Euler form. At infinity, divide by P^3 and put $t = 1/f$. The coefficients of δ and of the zeroth-order term are, respectively,

$$-\frac{12}{125}t + O(t^2), \quad \frac{6}{125}t + O(t^2),$$

while the normalized inhomogeneous term is

$$\frac{f(1 + 13f)}{P} = \frac{13}{125} + O(t).$$

Thus infinity is regular singular as well. The normalized inhomogeneous term is meromorphic at the other three singularities, so the usual Frobenius theorem applies to local branches of Y_c .

Let X be the complement of these four points, fix a branch of Y_c on a simply connected open subset of X , and let ρ denote the monodromy representation on V . For a loop γ , the branch difference $\kappa_\gamma = Y_c^\gamma - Y_c$ lies in V . These differences cannot all vanish: otherwise Y_c would be single-valued on X . Frobenius expansions at the four punctures then show that single-valuedness eliminates every nonintegral power and logarithmic term, so Y_c would extend meromorphically to $\mathbf{P}^1$, hence would be rational. This contradicts the preceding exclusion of a rational particular solution.

Consequently, the span $W = \langle \kappa_\gamma \rangle$ is nonzero. The cocycle identity $\kappa_{\alpha\beta} = \kappa_\alpha + \rho(\alpha)\kappa_\beta$ shows that W is monodromy-stable. Irreducibility of V therefore gives $W = V$, so the branch differences span the full three-dimensional homogeneous solution space.

Any inhomogeneous equation of order at most two would force all branch differences into a homogeneous solution space of dimension at most two. Together with the upper bound from (29), this proves that Y_c has exact inhomogeneous order three for every $c \in \mathbf{C}$.

Under the standing rationality assumption, $H = Y_{\eta_5}$ with $\eta_5 \in \mathbf{Q} \subset \mathbf{C}$. The definition of exact order now gives the $\mathbf{C}(f)$ -linear independence of $1, H, \delta H, \delta^2 H$, and hence their independence over $\mathbf{Q}(f)$. Finally, if $R = f(1 + 13f)P^2$, then the first-order operator $R\delta - (\delta R)$ annihilates R . Its composition with $\mathcal{L}_3$ is therefore a homogeneous operator annihilating H . Thus H , its derivatives, and 1 are holonomic, as required in Theorem 3.1. $\square$

5 Denominators and local continuation

Set $D_0 = 1$, and for $n \geq 1$ set $D_n = \text{lcm}(1, 2, \dots, n)$.

Lemma 5.1 (denominator rows). *Write*

$$A(q(f)) = \sum_{n \geq 0} \beta_n f^n, \quad A(q(f))G(q(f)) = \sum_{n \geq 0} \alpha_n f^n.$$

Then

$$\beta_n \in \mathbf{Z}, \quad D_n^3 \alpha_n \in \mathbf{Z}.$$

Under the rationality assumption, write $\eta_5 = a/b$, where $a, b \in \mathbf{Z}$ and $b > 0$. Then, for $r = 0, 1, 2$ and $n \geq 0$,

$$bD_n^3 [f^n] \delta^r H \in \mathbf{Z}.$$

Here $[f^n]$ denotes coefficient extraction.

Proof. By (3) and the integrality of $q(f)$,

$$B(f) := A(q(f)) \in \mathbf{Z}[[f]],$$

which proves the assertion for β_n . Write $G(q) = \sum_{m \geq 1} g_m q^m$. Every divisor d occurring in g_m satisfies $d \leq m$, hence $d \mid D_m$, so $D_m^3 g_m \in \mathbf{Z}$. Since $q(f)^m \in f^m \mathbf{Z}[[f]]$, we have

$$[f^n]G(q(f)) = \sum_{m=1}^n g_m [f^n] q(f)^m.$$

The bracketed coefficients are integers, and $D_m \mid D_n$ for $m \leq n$. It follows that $D_n^3 [f^n]G(q(f)) \in \mathbf{Z}$.

Now $\alpha_n = [f^n](B(f)G(q(f)))$. In its Cauchy product, the coefficient from degree s of $G(q(f))$ is cleared by D_s^3 , which divides D_n^3 whenever $s \leq n$; the coefficient from B is integral. Hence $D_n^3 \alpha_n \in \mathbf{Z}$. Finally, if $H = \sum_{n \geq 0} h_n f^n$, then

$$h_n = \alpha_n + \frac{a}{b} \beta_n, \quad [f^n] \delta^r H = n^r h_n \quad (r = 1, 2),$$

while the coefficient is h_n for $r = 0$. Thus $bD_n^3[f^n] \delta^r H$ is integral in all three cases, as claimed. $\square$

Lemma 5.2 (the 5-adic continuation disk). *The functions $H, \delta H, \delta^2 H$ are analytic on $|f|_5 < 125$.*

Proof. Normalize v_5 by $v_5(5) = 1$. We first identify the relevant wide open. From (21), if $t = v_5(f)$, the three terms of $N = 1 + 250f + 3125f^2$ have valuations

$$0, \quad 3 + t, \quad 5 + 2t.$$

For $t \geq 0$, the first is the unique minimum and $v_5(N) = 0$. For $-3 < t < -5/2$, the last is the unique minimum and $v_5(N) = 5 + 2t$. For $-5/2 < t < 0$, the first is again the unique minimum; at $t = -5/2$, possible cancellation can only increase $v_5(N)$. Finally, for $t \leq -3$, the last term is the unique minimum. Since $j = N^3/f$, this proves

$$v_5(j) = \begin{cases} -t, & t \geq 0, \\ > 0, & -3 < t < 0, \\ 15 + 5t, & t \leq -3. \end{cases} \quad (32)$$

including the strict inequality in the middle range. $\downarrow$

The unique supersingular j -invariant in characteristic five is 0: indeed, the Hasse invariant is $E_4 \bmod 5$, and $j = E_4^3/\Delta$ on the smooth locus. Consequently, $v_5(j) > 0$ is precisely the tube of the supersingular point, whereas the two ranges with $v_5(j) \leq 0$ give the two ordinary components and their cusp neighborhoods. Thus the ordinary component containing ∞ , the ordinary component containing 0, and the intervening supersingular annulus are, respectively,

$$|f|_5 \leq 1, \quad |f|_5 \geq 125, \quad 1 < |f|_5 < 125. \quad (33)$$

The Fricke relation (2) interchanges the first two regions and provides an independent check of their endpoints.

By the U_5 -calculation in Section 2, the overconvergent form $E_{-2}^{*(5)}$ has eigenvalue one and slope zero. Buzzard's analytic-continuation theorem [4] is stated for arbitrary integral weight and a nonzero U_5 -eigenvalue. It therefore applies in weight -2 and extends $E_{-2}^{*(5)}$ over the ordinary component containing ∞ together with the whole supersingular annulus.

For completeness, the harmless auxiliary-level step can be made explicit. Choose a sufficiently fine tame level prime to 5, pull the form back to that modular curve, and apply Buzzard's theorem there. Every deck transformation fixes the continued section: the two sections agree on the original overconvergent neighborhood, hence everywhere on the connected continuation domain by the rigid-analytic identity principle. The section therefore descends back to level $\Gamma_0(5)$; this is the auxiliary-level procedure described in [4]. By (33), the image of the resulting wide open on the coarse f -line is exactly the strict disk

$$|f|_5 < 125 \quad (34)$$

The classical form A is a global holomorphic section of weight two. Multiplication therefore produces the analytic weight-zero function H on this disk, agreeing with the original germ by its

q -expansion. Since f is a coordinate, d/df , and hence $\delta = f d/df$, preserves analyticity on the open disk. Thus δH and $\delta^2 H$ are analytic there as well. No continuation on the boundary $|f|_5 = 125$ is asserted. $\square$

Still under the standing rationality assumption, at a prime $\ell \neq 5$, Lemma 5.1 gives convergence of $H, \delta H, \delta^2 H$ on $|f| * \ell < 1$. Indeed, the fixed factor b does not affect the radius, and, for $n \geq 1$,

$$v * \ell(D_n) = \max\{e : \ell^e \leq n\} = \lfloor \log_\ell n \rfloor = O(\log n).$$

Thus the n -th coefficient of any of the three germs has ℓ -adic valuation at least $-3\lfloor \log_\ell n \rfloor - v_\ell(b)$, which is $o(n)$. At the complex place, the coefficients of $A(q)$ have polynomial growth, while the coefficient of q^n in $G(q)$ is at most $\sum_{d \geq 1} d^{-3} = \zeta(3)$. Hence both series are holomorphic for $|q| < 1$, and $H(f(q)) = A(q)(G(q) + \eta_5)$ is holomorphic there. Since $\delta = \theta/A$ after pullback to the q -disk, its f -Euler derivatives pull back meromorphically; their only possible additional poles occur at zeros of A .

6 Exact complex logarithmic energy

To avoid confusing the modular variable τ with its q -coordinate, write

$$\mathfrak{f}(q) = q \prod_{n \geq 1} \left(\frac{1 - q^{5n}}{1 - q^n} \right)^6$$

for the analytic q -expansion in (1). Take

$$\varphi_\infty(z) = \mathfrak{f}(z/2), \quad \varphi_5(z) = 5^{-3}z, \quad \varphi_\ell(z) = z \quad (\ell \neq 5). \quad (35)$$

All these maps vanish at the origin and have nonzero derivative there. The first is analytic for $|z| < 2$, because $\mathfrak{f}(q)$ is analytic for $|q| < 1$. At five,

$$|5^{-3}z|_5 = 125|z|_5 < 125 \quad (|z|_5 < 1),$$

so Lemma 5.2 applies. At every other finite place the identity map lands in the convergence disk established at the end of Section 5. That section also verifies the complex meromorphy required in Theorem 3.1.

Let

$$I = \iint_{\mathbf{T}^2} \log |\mathfrak{f}(z/2) - \mathfrak{f}(w/2)| d\mu(z) d\mu(w).$$

Lemma 6.1 (Modular Jensen energy). *Put $a = 5 \log 2 / (2\pi)$. Then*

$$I = -\log 2 + \frac{16\pi}{25} \arccos(a) - \frac{8 \log 2}{5} \sqrt{1 - a^2}. \quad (36)$$

Proof. Let $r = 1/2$, $y = \log 2 / (2\pi)$, and $q_u = e^{2\pi i(u+iy)} = re^{2\pi iu}$. We first justify the use of Fubini. The function $(z, w) \mapsto \mathfrak{f}(rz) - \mathfrak{f}(rw)$ is holomorphic on a neighborhood of the compact torus $\mathbf{T}^2$ and is not identically zero. Its pullback under $(u, v) \mapsto (e^{2\pi iv}, e^{2\pi iu})$ is therefore a nonzero real-analytic function on a neighborhood of $[0, 1] \times [0, 1]$. The logarithm of the absolute value of a nonzero real-analytic function is locally integrable (this follows, for example, from Weierstrass preparation and one-variable Fubini). Compactness consequently gives $(u, v) \mapsto \log |\mathfrak{f}(re^{2\pi iv}) - \mathfrak{f}(re^{2\pi iu})|$ in $L^1([0, 1] \times [0, 1])$, so Fubini's theorem applies.

For every u , the function $z \mapsto \mathfrak{f}(rz) - \mathfrak{f}(q_u)$ is holomorphic on a neighborhood of the closed unit disk, and its value at the origin is $-\mathfrak{f}(q_u) \neq 0$. The closed-circle form of Jensen's formula therefore gives

$$\int_0^1 \log |\mathfrak{f}(re^{2\pi i v}) - \mathfrak{f}(q_u)| dv = \log |\mathfrak{f}(q_u)| + \sum_{|z_0| < 1} \log \frac{1}{|z_0|}. \quad (37)$$

Here the zeros are counted with multiplicity. Zeros on the unit circle give integrable logarithmic singularities but no term in the sum. Because $\mathfrak{f}(q)/q$ is holomorphic and nonzero for $|q| < 1$, and has value one at $q = 0$, the mean of $\log |\mathfrak{f}(q)/q|$ on $|q| = r$ is zero. Hence

$$\int_0^1 \log |\mathfrak{f}(q_u)| du = \log r = -\log 2.$$

This is the contribution of the first term on the right of (37) after integration in u .

Since f is a degree-one coordinate on $X_0(5)$, we have $f(\tau') = f(\tau)$ for $\tau', \tau \in \mathfrak{H}$ exactly when $\tau' = \gamma\tau$ for some $\gamma \in \Gamma_0(5)$. For nonelliptic τ , the distinct q -coordinate preimages are therefore indexed by $\Gamma_\infty \backslash \Gamma_0(5)$. Representatives of these cosets, where

$$\Gamma_\infty = \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbf{Z} \right\},$$

are parametrized by primitive lower rows (c, d) with $5 \mid c$, modulo simultaneous sign. Indeed, these conditions are necessary and sufficient for (c, d) to be the lower row of an element of $\Gamma_0(5)$, and two such matrices with the same lower row differ by left multiplication by an element of Γ_∞ . For $\tau = u + iy$, the corresponding zero is $z_0 = r^{-1}e^{2\pi i \gamma \tau}$. Since $\text{Im}(\gamma\tau) = y/|c\tau + d|^2$, it lies in $\mathbb{D}$ exactly when

$$|c\tau + d| < 1, \quad (38)$$

and its Jensen weight is

$$2\pi y (|c\tau + d|^{-2} - 1). \quad (39)$$

Choose the simultaneous sign so that $c \geq 0$. For $c = 0$, primitivity gives $d = 1$; this is the identity coset, whose zero $z_0 = e^{2\pi i u}$ is on the boundary and has Jensen weight zero. For $c > 0$, the inequality $|c|y < 1$ is necessary. The elementary bounds $2/3 < \log 2 < 1$ and $3 < \pi < 22/7$ give $5 < 1/y < 10$. Since c is a positive multiple of five, the only possible nonidentity value is $c = 5$.

Put

$$a = 5y = \frac{5 \log 2}{2\pi}, \quad b = \sqrt{1 - a^2}.$$

The preceding bounds give $0 < a < 1$. For $c = 5$, condition (38) becomes $|5u + d| < b < 1$. As $0 \leq u < 1$, the only possible integers are $d = 0, -1, \dots, -5$; primitivity excludes 0 and -5 . Thus the complete nonidentity list is $d = -1, -2, -3, -4$. Parameters u for which $\tau = u + iy$ represents an elliptic point of $X_0(5)$ form a finite set, hence a null set, and do not affect the integral. Boundary coincidences likewise do not affect it, because their Jensen weight is zero.

For $d = -j$, the interior condition is

$$\frac{j - b}{5} < u < \frac{j + b}{5}.$$

All four intervals lie in $(0, 1)$. They may overlap, but Jensen's formula sums zeros with multiplicity, so the four coset contributions must be added separately. With $t = 5u - j$, each contribution has

Jacobian $du = dt/5$. Integrating (39) therefore gives

$$\begin{aligned} I &= -\log 2 + \frac{8\pi y}{5} \int_{-b}^b \left(\frac{1}{t^2 + a^2} - 1 \right) dt \\ &= -\log 2 + \frac{8\pi y}{5} \left(\frac{2}{a} \arctan \frac{b}{a} - 2b \right). \end{aligned}$$

Since $a^2 + b^2 = 1$ and $a, b > 0$, $\arctan(b/a) = \arccos(a)$. Substitution of $a = 5y$ and $2\pi y = \log 2$ now gives

$$I = -\log 2 + \frac{16\pi}{25} \arccos(a) - \frac{8 \log 2}{5} b,$$

which is (36). □

7 Completion of the proof

Lemma 7.1 (rational numerical certificate). *With I as in Lemma 6.1,*

$$I < \frac{183}{500} < \frac{231}{500} < -4 \log 2 + 9 \log 5 - \frac{45}{4},$$

and

$$-\log 2 + 3 \log 5 - \frac{45}{16} > \frac{33}{25}.$$

Proof. We include a certificate involving rational arithmetic only. For $0 < z < 1$, set

$$S_N(z) = 2 \sum_{k=0}^{N-1} \frac{z^{2k+1}}{2k+1}.$$

Termwise integration of the geometric series gives

$$S_N(z) < \log \frac{1+z}{1-z} < S_N(z) + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}. \quad (40)$$

We use $z = 1/3$, $N = 35$ for $\log 2$, and $z = 1/9$, $N = 18$ for $\log(5/4)$. For $0 < x \leq 1$, put

$$T_M(x) = \sum_{k=0}^{M-1} \frac{(-1)^k x^{2k+1}}{2k+1}.$$

The alternating-series remainder gives, for every positive even M ,

$$T_M(x) < \arctan x < T_{M+1}(x). \quad (41)$$

We use $M = 60$ at $x = 1/5$ and $M = 24$ at $x = 1/239$ in Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. (Rational tangent addition gives tangent one. The inequalities $x - x^3/3 < \arctan x < x$, together with $\pi > 3$, show that the angle lies strictly between 0 and $\pi/2$, which fixes the branch.) Finally, we use $M = 200$ to bound the arctangent in

$$\arccos a = \frac{\pi}{2} - \arctan \left(\frac{a}{\sqrt{1-a^2}} \right).$$

This gives the following strict rational intervals.

quantity	strict lower bound	strict upper bound
$\log 2$	$693147180559945/10^{15}$	$693147180559946/10^{15}$
$\log(5/4)$	$223143551314209/10^{15}$	$223143551314210/10^{15}$
π	$3141592653589793/10^{15}$	$3141592653589794/10^{15}$
a	$551589000381628/10^{15}$	$551589000381629/10^{15}$
$\sqrt{1-a^2}$	$834116043879983/10^{15}$	$834116043879984/10^{15}$
$\arccos a$	$986528275021999/10^{15}$	$986528275022000/10^{15}$

For clarity, the table records outward decimal roundings, not the intermediate endpoints used recursively. We now specify those exact endpoints. Let $L_- < \log 2 < L_+$ be the two rational bounds from (40) with $(z, N) = (1/3, 35)$, and put

$$\begin{aligned} P_- &= 16T_{60}(1/5) - 4T_{25}(1/239), \\ P_+ &= 16T_{61}(1/5) - 4T_{24}(1/239), \\ A_- &= \frac{5L_-}{2P_+}, & A_+ &= \frac{5L_+}{2P_-}. \end{aligned}$$

Thus $P_- < \pi < P_+$ and $A_- < a < A_+$. With $R = 10^{30}$, define

$$\begin{aligned} B_- &= R^{-1} \left[R\sqrt{1 - A_+^2} \right], \\ B_+ &= R^{-1} \left(\left[R\sqrt{1 - A_-^2} \right] + 1 \right). \end{aligned}$$

Integer squaring verifies these floor values and gives $B_- < b < B_+$. Set $t_- = A_-/B_+$ and $t_+ = A_+/B_-$; exact cross-multiplication gives

$$0 < t_- < \frac{a}{b} < t_+ < 1.$$

Finally, define the rational numbers

$$C_- = \frac{P_-}{2} - T_{201}(t_+), \quad C_+ = \frac{P_+}{2} - T_{200}(t_-).$$

Equation (41) gives $C_- < \arccos a < C_+$. Exact cross-multiplication places each of these exact endpoints strictly inside its displayed decimal interval. Thus every entry in the table is certified by rational arithmetic; no rounded endpoint is reused to derive a later row.

Writing $X_{\text{tab},-}$ and $X_{\text{tab},+}$ for the displayed endpoints of a quantity X , put

$$U_{\text{tab}} = -(\log 2) * \text{tab}, - + \frac{16}{25} \pi * \text{tab}, + (\arccos a) * \text{tab}, + \\ - \frac{8}{5} (\log 2) * \text{tab}, - b_{\text{tab}}, -.$$

The signs in (36) now give

$$I < U_{\text{tab}} = \frac{45665164124379924046682621253}{1250000000000000000000000000000} < \frac{183}{500}.$$

Since $\log 5 = 2 \log 2 + \log(5/4)$, the two remaining expressions are

$$\begin{aligned} -4 \log 2 + 9 \log 5 - \frac{45}{4} &= 14 \log 2 + 9 \log(5/4) - \frac{45}{4}, \\ -\log 2 + 3 \log 5 - \frac{45}{16} &= 5 \log 2 + 3 \log(5/4) - \frac{45}{16}. \end{aligned}$$

Their lower endpoints give, respectively,

$$-4 \log 2 + 9 \log 5 - \frac{45}{4} > \frac{462352489667111}{10^{15}} > \frac{231}{500}$$

and

$$-\log 2 + 3 \log 5 - \frac{45}{16} > \frac{82666659796397}{62500000000000} > \frac{33}{25}.$$

□

i

Proof of Theorem 1.1. Assume for contradiction that $\zeta_5(3) \in \mathbf{Q}$. Then $\eta_5 \in \mathbf{Q}$, and all four germs

$$F_1 = 1, \quad F_2 = H, \quad F_3 = \delta H, \quad F_4 = \delta^2 H$$

belong to $\mathbf{Q}[[f]]$. Equation (26) expresses $\delta^3 H$ as a $\mathbf{Q}(f)$ -linear combination of $1, H, \delta H$. Hence the $\mathbf{Q}(f)$ -span of $1, H, \delta H, \delta^2 H$ is stable under δ , so each of its elements is holonomic. Proposition 4.3 says that the four displayed germs are linearly independent over $\mathbf{Q}(f)$.

We next verify the denominator hypothesis of Theorem 3.1. Write $\eta_5 = a_0/b_0$ in lowest terms with $b_0 > 0$. Since $D_n = \Lambda(n)$, Lemma 5.1 shows that the common multiplier $C = b_0$ in (7) works with the capping matrix

$$\mathbf{b} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}. \quad (42)$$

Indeed, the first row requires no denominator, while each coefficient in the last three rows becomes integral after multiplication by $b_0 D_n^3$. Thus $m = 4$, and each column has $u_j = b_j = 1$. Therefore

$$\tau(\mathbf{b}) = 3 - \frac{3}{16} = \frac{45}{16}. \quad (43)$$

It remains to check the local hypotheses and compute the two adelic quantities. The maps in (35) are analytic on neighborhoods of the corresponding closed unit disks, vanish simply at the origin, and are the identity at all primes other than five. At five, $\varphi_5(|z| * 5 < 1)$ lies in the continuation disk of Lemma 5.2; at every $\ell \neq 5$, the coefficient estimate following that lemma gives meromorphic (in fact analytic) continuation on the open unit disk. At the complex place, the last paragraph of Section 5 gives the required meromorphic continuation after composition with $\varphi * \infty$. Thus all local hypotheses of Theorem 3.1 hold for the four germs.

For the maps in (35), normalized by $|5| * 5 = 5^{-1}$,

$$\mathcal{A} = I + 3 \log 5, \quad (44)$$

$$\mathcal{L} = -\log 2 + 3 \log 5. \quad (45)$$

Indeed, $f'(0) = 1$, so $\varphi * \infty'(0) = 1/2$. At five, both the derivative and the boundary supremum contribute $\log |5^{-3}|_5 = 3 \log 5$; all other finite places contribute zero. Put

$$\mathcal{D} = \mathcal{L} - \tau(\mathbf{b}) = -\log 2 + 3 \log 5 - \frac{45}{16}.$$

Lemma 7.1 gives $\mathcal{D} > 33/25 > 0$. Theorem 3.1 therefore gives

$$4 \leq \frac{I + 3 \log 5}{-\log 2 + 3 \log 5 - 45/16}. \quad (46)$$

The first chain in Lemma 7.1 gives $I < -4 \log 2 + 9 \log 5 - 45/4$. Equivalently,

$$4\mathcal{D} - (I + 3 \log 5) = -4 \log 2 + 9 \log 5 - \frac{45}{4} - I > 0.$$

Since $\mathcal{D} > 0$, this proves

$$\frac{I + 3 \log 5}{-\log 2 + 3 \log 5 - 45/16} < 4. \quad (47)$$

Equations (46) and (47) contradict each other. The rationality assumption was therefore false. Since $\zeta_5(3) \in \mathbf{Q}_5$ by construction of the Kubota–Leopoldt value, Theorem 1.1 follows. $\square$

Comparison with the elementary level-five estimate

Calegari’s elementary level-five approximants have exponent

$$\frac{3}{3/2 + 3/\log 5} = 0.891794 \dots < 1,$$

and therefore do not prove Theorem 1.1. The argument above uses both the finite-place holonomy contribution and the optimized row-wise invariant $45/16$. If that invariant is replaced by the unoptimized value 3, the resulting quotient is $4.575 \dots$, which yields no contradiction.

Role of computation and reproducibility

The scripts are publicly available in the github repository github.com/shivakintali/zeta-5-3. The script `verify_level5_ode.py` checks through degree 80 the two Hauptmodul identities, the denominator rows, and (26); it also verifies exactly the polynomial factorizations, the differentiated rational identity, and the indicial polynomial used in Section 4. The script `energy_exact.py` checks the rational interval inequalities used in the coset enumeration of Section 6; it also verifies every rational endpoint and every displayed inequality in Lemma 7.1.

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